Year 9 · Chapter 10 · Surface area and volume: answers (teacher copy)

10.1 Metric units (booklet p. 3)

WE DO examples (teacher)

1. Ex 1a: 8 × 1000 = 8 000 m
2. Ex 1b: 250 ÷ 100 = 2.5 m
3. Ex 1c: 3.5 × 1000 = 3 500 g
4. Ex 1d: 4500 ÷ 1000 = 4.5 L
5. Ex 2a: 3.5 × 60 = 210 min
6. Ex 2b: 150 ÷ 60 = 2.5 min
7. Ex 2c: 4.2 × 1000 = 4 200 MB
8. Ex 2d: 2 500 000 ÷ 1 000 000 = 2.5 m

Easy

1. 6 000 m
2. 3 km
3. 400 cm
4. 7 m
5. 90 mm
6. 12 cm
7. 2 500 m
8. 0.45 m
9. 5 000 g
10. 2 kg
11. 3 000 kg
12. 6.5 t
13. 800 g
14. 0.75 kg
15. 4 g
16. 1 200 kg
17. 7 000 mL
18. 3 L
19. 2 000 L
20. 4.5 kL
21. 1 500 mL
22. 0.25 L
23. 600 L
24. 0.08 L
25. 400 000 cm
26. 3 000 mm
27. 0.6 km
28. 5 m
29. 2 000 000 mg
30. 1 000 000 mL
31. 50 000 cm
32. 0.08 m

Medium

1. 300 s
2. 240 min
3. 3 min
4. 150 min
5. 1.5 h
6. 72 h
7. 4 s
8. 90 s
9. 3 000 MB
10. 4.5 GB
11. 2 400 GB
12. 34 kB
13. 2 600 000 m
14. 700 ms
15. 3 m
16. 5 000 µs
17. 80 videos
18. 2 h 15 min

Mixed practice

1. 750 m (10.1)
2. 40 cm² (Year 8)
3. 1.2 m (= 120 cm) (10.1)
4. 3.5 (Year 8)
5. 7 200 s (10.1)
6. 28 m (Year 8)
7. 5 cups (10.1)
8. 2 500 MB (10.1)

10.2 Precision and absolute error (booklet p. 5)

WE DO examples (teacher)

1. Ex 1a: precision 1 mm, absolute error 0.5 mm
2. Ex 1b: precision 50 mL, absolute error 25 mL
3. Ex 1c: precision 1 m, absolute error 0.5 m
4. Ex 1d: precision 0.1 kg, absolute error 0.05 kg
5. Ex 2a: precision 0.01 m, absolute error 0.005 m
6. Ex 2b: 2.445 m to 2.455 m
7. Ex 2c: 161.5 cm to 162.5 cm
8. Ex 2d: 625 mL to 675 mL

Easy

1. 1 m
2. 0.1 cm
3. 0.01 kg
4. 1 mm
5. 0.1 L
6. 0.01 t
7. 1 g
8. 0.1 s
9. 0.5 m
10. 0.05 kg
11. 0.5 mL
12. 0.005 m
13. 0.5 cm
14. 0.05 L
15. 0.005 s
16. 0.5 g
17. precision 0.1 kg, absolute error 0.05 kg
18. precision 20 mL, absolute error 10 mL
19. precision 2 °C, absolute error 1 °C
20. precision 0.5 m, absolute error 0.25 m

Medium

1. 0.005 kg
2. 0.05 m
3. 0.05 L
4. 0.0005 km
5. 0.005 g
6. 0.005 m
7. 0.05 cm
8. 0.0005 t
9. 23.5 m to 24.5 m
10. 6.35 cm to 6.45 cm
11. 152.5 g to 153.5 g
12. 3.245 kg to 3.255 kg
13. 9.75 s to 9.85 s
14. 71.5 kg to 72.5 kg
15. 1.55 L to 1.65 L
16. 0.445 m to 0.455 m
17. 245 g to 255 g
18. No: the limits are 12.45 m to 12.55 m, and 12.56 m is longer than 12.55 m

Mixed practice

1. 0.01 m (10.2)
2. 3 200 m (10.1)
3. 0.5 mm (10.2)
4. 3 (Year 8)
5. 17.5 cm to 18.5 cm (10.2)
6. 0.75 L (10.1)
7. 10 g (10.2)
8. 0.05 L (10.2)

10.3 Percentage error (booklet p. 7)

WE DO examples (teacher)

1. Ex 1a: 0.5 ÷ 20 × 100 = 2.5%
2. Ex 1b: 0.5 ÷ 4 × 100 = 12.5%
3. Ex 1c: 0.05 ÷ 2.5 × 100 = 2%
4. Ex 1d: 0.5 ÷ 50 × 100 = 1%
5. Ex 2a: 0.5 m; 0.5 ÷ 24 × 100 = 2.08%
6. Ex 2b: 0.005 ÷ 3.25 × 100 = 0.15%
7. Ex 2c: 12.5 ÷ 375 × 100 = 3.33%
8. Ex 2d: b: it has the smallest percentage error (0.15%)

Easy

1. 5%
2. 2%
3. 1%
4. 12.5%
5. 2.5%
6. 0.5%
7. 10%
8. 1.25%
9. 2%
10. 10%
11. 3.125%
12. 0.4%
13. 25%
14. 12.5%
15. 0.08%
16. 6.25%
17. 7.1%
18. 3.3%
19. 5.6%
20. 1.7%
21. 1.4%
22. 8.3%
23. 1.1%
24. 2.2%

Medium

1. 2.08%
2. 1.39%
3. 0.04%
4. 0.29%
5. 6.25%
6. 0.21%
7. 1.04%
8. 0.53%
9. 1.43%
10. 16.67%
11. 2.63%
12. 2.08%
13. height: 0.30% is smaller than 3.33%
14. 0.40%

Mixed practice

1. 6.25% (10.3)
2. 5.25 kg to 5.35 kg (10.2)
3. 2.5% (10.3)
4. 4.5 kg (10.1)
5. 10% (10.3)
6. 100 m (0.5%, not 5%) (10.3)
7. 30 cm² (Year 8)
8. 2% (10.3)

10.4 Perimeter and area of composite shapes (booklet p. 9)

WE DO examples (teacher)

1. Ex 1a: missing sides 6 m and 5 m; P = 10 + 4 + 6 + 5 + 4 + 9 = 38 m
2. Ex 1b: 10 × 4 + 4 × 5 = 40 + 20 = 60 m²
3. Ex 1c: 8 × 5 + ½ × 8 × 3 = 40 + 12 = 52 cm²
4. Ex 1d: 10 × 6 − 3 × 2 = 60 − 6 = 54 m²
5. Ex 2a: the steps add to 12 m across and 9 m up: P = 2 × (12 + 9) = 42 m
6. Ex 2b: 12 × 3 + 8 × 3 + 4 × 3 = 72 m²
7. Ex 2c: 7.5 × 4 + ½ × 7.5 × 2.4 = 30 + 9 = 39 cm²
8. Ex 2d: 12 × 8 − ½ × 5 × 4 = 96 − 10 = 86 m²

Easy

1. 30 cm
2. 44 m
3. 44 cm
4. 46 m
5. 36 cm²
6. 50 m²
7. 33 cm²
8. 55 m²
9. 48 m²
10. 39 cm²
11. 42 m²
12. 45 cm²

Medium

1. 36 cm
2. 37 m
3. 28 m
4. 40 cm
5. 48 m²
6. 66 cm²
7. 85 m²
8. 64 cm²

Mixed practice

1. 34 m (10.4)
2. 1.25% (10.3)
3. 90 m² (10.4)
4. 240 cm (10.1)
5. 18 m² (10.4)
6. 7.15 m to 7.25 m (10.2)
7. 18 m² (10.4)
8. 28 cm (10.4)

10.5 Areas of quadrilaterals (booklet p. 11)

WE DO examples (teacher)

1. Ex 1a: A = bh = 12 × 4 = 48 cm²
2. Ex 1b: A = ½(a + b)h = ½ × (14 + 8) × 6 = 66 cm²
3. Ex 1c: A = ½xy = ½ × 10 × 6 = 30 m²
4. Ex 1d: A = ½xy = ½ × 8 × 12 = 48 cm²
5. Ex 2a: the 5 cm side is not needed: A = 12 × 4 = 48 cm²
6. Ex 2b: ½ × (7.5 + 4.5) × 3.2 = 19.2 m²
7. Ex 2c: 54 = 9 × h, so h = 54 ÷ 9 = 6 cm
8. Ex 2d: ½ × 1.2 × 0.8 = 0.48 m²

Easy

1. 40 cm²
2. 48 m²
3. 105 cm²
4. 45 m²
5. 32 cm²
6. 50 m²
7. 42 cm²
8. 36 m²
9. 24 cm²
10. 30 m²
11. 42 cm²
12. 72 m²

Medium

1. 36 m²
2. 40 cm²
3. 96 cm²
4. 40.5 m²
5. 8 cm
6. 6 m
7. 5 cm
8. 50 cm²

Mixed practice

1. 21 m² (10.5)
2. 26 cm (10.4)
3. 25 cm² (10.5)
4. 2% (10.3)
5. 18 m² (10.5)
6. 600 m (10.1)
7. 35 cm² (10.5)
8. 5 m (10.5)